

Round Numbers, Volatility, and Investor Identity in a Bull Market: Evidence from the Korean Stock Index, 2025–2026

Nakcho Choi¹ and Youngchul Kim^{*2}

¹Ph.D. in Applied Physics; Principal Engineer, Samsung Display

²Department of Economics, Sogang University, Seoul, Korea

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Abstract

We examine whether 1,000-point round numbers act as price barriers in the Korea Composite Stock Price Index (KOSPI) during the exceptional bull market of 2025–2026, when the index rose roughly 2.8-fold (2,771 to 8,161) over 244 trading days. While the established literature on *individual-security* price clustering attributes the round-number effect primarily to retail investors' cognitive bias (Bloomfield et al., 2025), we argue that at the *index* level the mechanism inverts: index round numbers are dominated by exchange-traded-fund (ETF) flows, index-futures arbitrage, and program trading, so that institutional and foreign

^{*}Corresponding author. Department of Economics, Sogang University, Seoul, Korea. [*Placeholder—to be confirmed. Co-authorship requires Prof. Kim's prior review and consent before any submission.*]

investors—not retail—drive selling at thresholds. Applying the standard psychological-barrier methodology (Aggarwal and Lucey, 2007)—uniformity, barrier-proximity, barrier-hump, and conditional-effect tests on M -values—we find that the trailing-digit distribution of closing levels is strongly non-uniform (Kolmogorov–Smirnov $p < 0.001$), but that the canonical barrier-proximity and conditional-variance tests are insignificant. Crucially, a volatility-clustering effect that is significant under a *relative-distance* definition (Mann–Whitney $p = 0.027$) vanishes under the standard *absolute trailing-digit* (M -value) definition ($p = 0.58$). This definition sensitivity, which arises because a fixed percentage band maps to a moving point band during a rapid ascent, is itself a methodological contribution for studying round-number effects in trending markets. In documented breakdown episodes, foreign and institutional investors were the net sellers (via program trading) while retail investors absorbed the flow—consistent with Korean evidence that institutional and foreign herding stabilizes while retail herding destabilizes prices (Kim et al., 2020), and with the informed-trading view of index-threshold behavior.

Keywords: psychological barriers; round numbers; price clustering; investor type; program trading; Korean stock market

JEL: G12, G14, G15, G41

1 Introduction

Whether salient “round numbers” act as psychological barriers in asset prices is a long-standing question in behavioral finance (Donaldson and Kim, 1993; Aggarwal and Lucey, 2007). The Korean market in 2025–2026 offers an unusual natural experiment: propelled by an AI semiconductor super-cycle, the KOSPI rose from about 2,700 to nearly 8,800 within a year, crossing six 1,000-point levels in rapid succession. Each crossing was popularly described in terms of “resistance” at the round number, and program-trading sidecars were triggered at the 5,000 breakdown (2 February 2026) and the 8,000 threat (5 June 2026).

We ask three questions. First, do 1,000-point levels behave as barriers under the standard methodology? Second, how sensitive are the conclusions to how “proximity” is defined in a rapidly trending index? Third, which investor types drive selling at these levels? Our answers

are deliberately measured: the evidence for a barrier is *mixed and definition-dependent*, and we regard the documentation of that sensitivity—together with the investor-identity pattern specific to Korea—as the paper’s contribution.

A central theme organizes our investor-identity analysis: a *scale inversion* between individual securities and the aggregate index. In individual stocks, round-number clustering is well documented as a retail-driven cognitive phenomenon (Bloomfield et al., 2025; ?). At the index level, however, the tradable instruments are different—index ETFs, index futures, and basket program trades—and these are overwhelmingly the province of institutional and foreign investors. We therefore hypothesize that the investor identity behind round-number selling *flips* from retail (at the single-stock level) to institutional/foreign (at the index level). This conjecture is consistent with Korean high-frequency evidence that institutional and foreign herding reduces realized volatility (informed trading), whereas retail herding raises it (noise trading) (Kim et al., 2020).

2 Data and Methodology

2.1 Data

Daily KOSPI closing levels (4 June 2025–5 June 2026; 244 trading days) are obtained via FinanceDataReader. We compute log returns r_t . Following the barrier literature, we define the M -value as the trailing digits of the level, $M_t = \lfloor (P_t \bmod 1000)/10 \rfloor \in \{0, \dots, 99\}$, so that a barrier at a 1,000-point level corresponds to M near 00. We also use a *relative-distance* measure $d_t = (P_t - R_t)/R_t \times 100$, where R_t is the nearest 1,000 multiple. Investor-type net flows and sidecar events are taken from public reports.

2.2 Tests

We implement the Aggarwal–Lucey battery: (i) a uniformity test (Kolmogorov–Smirnov and χ^2) on the M -value distribution; (ii) a barrier-proximity test, regressing the frequency $f(M)$ on a barrier dummy; (iii) a barrier-hump test, $f(M) = a + bM + cM^2$; and (iv) conditional-effect tests comparing mean returns and variances near versus far from barriers, including a GARCH(1,1)

specification. We additionally test the absolute-return distribution under both the M -value and relative-distance definitions of proximity.

3 Results

3.1 Uniformity

The M -value distribution is strongly non-uniform (KS $p < 0.001$; χ^2 $p < 0.001$; Table 1), confirming that closing levels cluster rather than distribute evenly across trailing digits—a necessary but not sufficient condition for barriers (Ley and Varian, 1994).

3.2 Barrier and conditional-effect tests

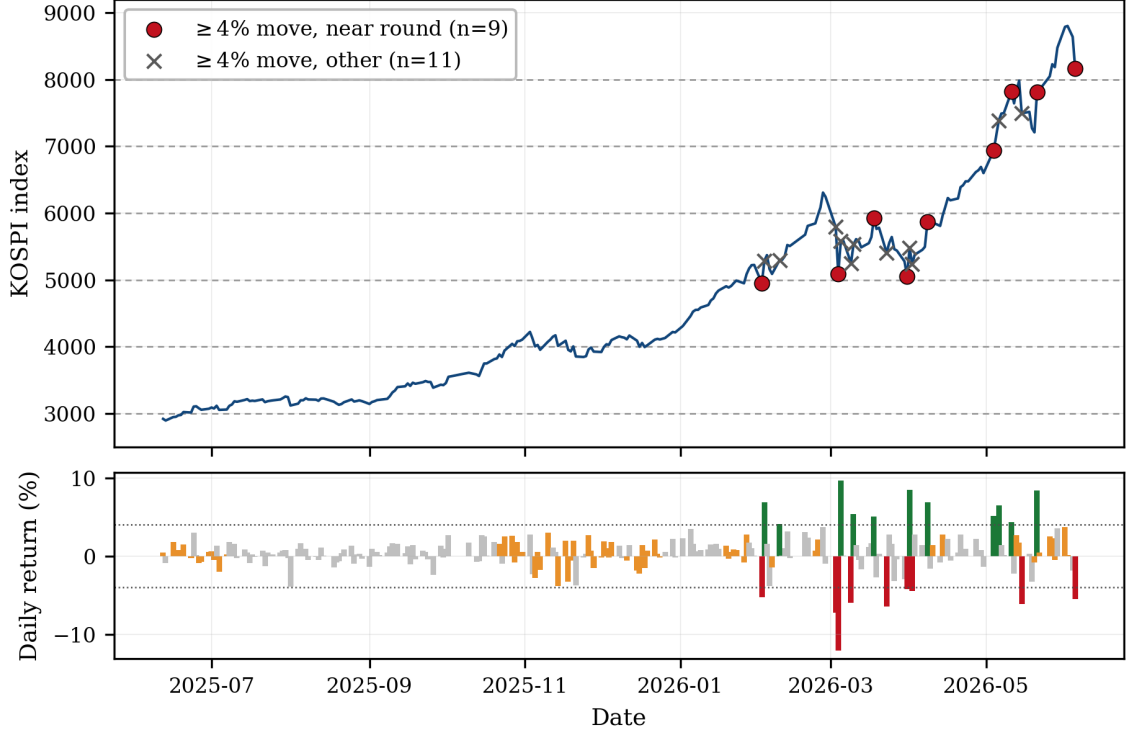
The barrier-proximity coefficient is not significant under either a strict (00 ± 2) or wide ($95-05$) window, and the barrier-hump quadratic term is small. Conditional-effect tests show no significant difference in mean returns near barriers, and the GARCH conditional volatility is, if anything, slightly *lower* near M -value barriers. Thus the canonical barrier tests do not support a barrier under the absolute trailing-digit definition.

3.3 Definition sensitivity (key result)

The contrast in Table 1 is the paper’s core empirical finding. Under the relative-distance definition ($|d_t| \leq 3\%$), near-round windows exhibit significantly larger absolute returns (Mann–Whitney $p = 0.027$; volatility ratio 1.22). Under the absolute M -value definition (M within ± 3 of 00), the same test is insignificant ($p = 0.58$; ratio 0.54). The divergence is mechanical: during a 2.8-fold ascent, a fixed 3% band spans a *moving* number of index points (about 80 points near 2,700 but 240 near 8,000), so the two definitions select different observation sets. Researchers studying round-number effects in strongly trending markets must therefore state and justify the proximity definition; conclusions are not robust to it.

Table 1: Barrier tests under two proximity definitions (KOSPI, 244 days).

Test	Statistic	Verdict
Uniformity (KS)	$p < 0.001$	non-uniform (clustering)
Uniformity (χ^2 , 10 bins)	$p < 0.001$	non-uniform
Barrier proximity (00 ± 2)	$\beta > 0, p = 0.54$	n.s.
Barrier hump (M^2 term)	small	weak
Cond. variance near barrier (M -value)	ratio 0.54, $p = 0.58$	n.s.
Volatility, relative distance ($\pm 3\%$)	ratio 1.22, $p = 0.027$	significant

**Figure 1:** KOSPI closing series (top) and daily returns (bottom). Red circles mark $\geq 4\%$ moves in relative-distance near-round windows; gray crosses mark other $\geq 4\%$ moves. Dashed lines are 1,000-point levels.

3.4 Investor identity at breakdowns

Table 2 summarizes the two sidecar episodes. At the 5,000 breakdown and the 8,000 threat, foreign and institutional investors were net sellers while retail investors were net buyers. This pattern is the empirical counterpart of our scale-inversion hypothesis: whereas single-stock round-number clustering is retail-driven (Bloomfield et al., 2025), index-level threshold selling is dominated by institutional and foreign program flows. It aligns with Kim et al. (2020), who show using Korea Exchange high-frequency data that institutional and foreign herding is information-driven and volatility-reducing while retail herding is noise-driven and volatility-increasing, and

with the broader informed-trading view of index thresholds (Chen, 2018; Yu et al., 2024). We treat the two episodes as suggestive case evidence; a full-sample test with transaction-level, investor-type net flows is left for future work.

Table 2: Investor net flows at round-number breakdowns (public reports).

Date	Event	Foreign	Institution	Retail
2026-02-02	5,000 breakdown, sell sidecar (-5.3%)	net sell	net sell	net buy
2026-02-03	rebound, buy sidecar ($+6.8\%$)	return	return	take profit
2026-06-05	8,000 threat (-5.5%)	net sell	net sell	net buy

4 Discussion and Conclusion

Our findings are intentionally cautious. The KOSPI’s closing levels cluster (uniformity is rejected), but the canonical barrier tests do not establish a psychological barrier under the standard M -value definition. A volatility-clustering effect appears under a relative-distance definition but not under the absolute one; we attribute this to the moving mapping between percentage and point bands in a fast-rising index, and we flag it as a methodological caution for trending-market studies. The investor-identity evidence, though based on representative episodes, points to foreign and institutional program selling at thresholds, with retail absorption—a pattern aligned with the institution-driven, informed-trading branch of the literature and with Korea’s distinctive microstructure.

Three extensions would sharpen these results: (i) transaction-level investor-type flows to test the selling-identity pattern across the full sample; (ii) cross-market replication (other Asia-Pacific indices, prior bull runs) to assess external validity; and (iii) intraday data to identify the causal direction between threshold proximity and order flow.

Author Contributions and Disclosure

N.C. designed the study, collected data, and performed the analysis. The corresponding-author role for Y.C.K. is provisional and subject to his review and consent. This draft is for academic

discussion and is not investment advice.

Data and Code Availability

The KOSPI closing series and all Python analysis scripts reproducing the tables and figures are available from the authors on request.

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